

Addiction vs. Reflection: Unlocking the Potential of Games

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I was just ten years old when the circumstances of life led me to lock-picking for the first time.

No, I wasn't starving, nor had I been abducted by a band of thieves. I was driven by something which, at the time, seemed much more urgent.

You see, my mother had lost her patience in trying to compete for our attention with the Nintendo. So, having determined that self-regulation with these newfangled computer games was impossible, she resorted to placing a lock on the power cable to prevent unauthorized access.

We were devastated.

Didn't she understand that the fate of an entire fantasy world was at stake and that her kids alone had been called upon to save it? Could she not fathom the wrenching heartache that ensued when we were prevented from fulfilling our virtual calling in life for hours at a time?

But, when our outrage fell on deaf ears, we returned to what games had already taught us about solving problems. We searched high and low for the key, but to no avail. We tried a clothespin, no luck there either. Finally, we fell back on brute force and, with the help of a screwdriver, we managed to wrench the lock open and, unbeknownst to our mother, return to blissfully saving the world.

The Magic of Narrative and Imagination

Two decades have gone by and I now find myself a parent (a.k.a. reformed gamer) looking at the same picture, but through the opposite side of the telescope.

I watch as my own boys lose themselves in fantasy adventures, digital puzzles and immersive simulations, and it is now me who loses my patience when competing with the screen for attention or trying to promote some non-digital activity, only to be told that everything else in life seems boring by comparison.

Yet the empathetic part of me can't grow too frustrated at them, knowing all too well what it is to be held in thrall to the siren calls of the Gameworld.

I remember as well that I often felt as though I was learning something while playing, or at least my mind was doing a lot of work. So I've withheld judgment for the time being to see whether there might in fact be something educationally redeeming about games for this generation.

Looking at them through a more critical eye, the first thing I've found is that games turn many older notions of education on their collective head.

After all, where education has, in times past, been characterized as a deliberate confrontation with boredom, games manage to tap into the most primal cues of narrative and the imagination. This enables them to take otherwise dry and abstract information and rearrange it into compelling and memorable systems that leave learners wanting more.

For instance when I studied cell biology in the 12th grade with terms like DNA, rNa, ribosomes, lysosomes, membranes, mitochondria, ATP and glucose, I read a textbook with definitions lined up one after another in academic speak and the occasional picture to accompany the text. The only context at the time was that it was Chapter 11 stuck between Chapters 10 and 12, all of which were filled with a bunch of information I would need to digest in order to pass the big exam.

When my kids learned cell biology (they were six and eight at the time), they simply played the game [Cellcraft](#) for a few weeks. In the game, they took control of a single-celled organism, planted in space as the only hope for continuing a race of Platypuses whose planet was destroyed. They had to hunt down glucose to survive; they watched the mitochondria in action turning fuel into energy; and they quickly learned the names and purposes of every biological defense mechanism at their disposal as they fended off waves of attacking viruses seeking to compromise the host cell. This was serious stuff, dripping with context, and within a comparatively short time these boys knew their way around a cell better than I ever had.

The secret wasn't just graphics or moving pictures. The secret was that they cared.

Taking an Active Role

Games find ways to make us care . . . a lot.

I found the same thing when I spent a summer as an intern on the [Making History](#) project at [Muzzy Lane](#).

I had never developed a taste for history in school, as it seemed frozen and irrelevant. Yet here I was working on a WWII history simulation that made Axis and Allies look like child's play. Every aspect of government, diplomacy, the economy, policy-making, military strategy and history itself was at your fingertips. You could follow history, try to change it, figure out why things had happened and what might have been done differently if different choices had been made. This wasn't dust-covered history; it was living, engaging stuff.

Simulations and games allow us to flip the switches, call the shots and live through the consequences of our decisions for good and bad. We experience the elation of success and the agony of defeat as we experiment and learn with dogged tenacity until we overcome the challenge at hand.

Outside the world of simulations, I've found games to be adept at the principle of compete and repeat.

My seven-year-old, who hadn't shown a lot of interest in traditionally taught writing or math, got onto a skill-building site a few months ago. As soon as it was clear to him that he was in competition with kids from around the globe, many of them twice his age, he couldn't be kept from drilling the same exercises over and over and over again until he could win the races. Within a few weeks, he had his arithmetic facts down pat: addition, subtraction, multiplication and division. He could type a list of spelling words just as quickly as his parents, and on most of the games he could beat us hands down. The same thing happened with geography when our kids discovered [Stack the Countries](#) on the iPod, and within a couple of weeks could pretty much pinpoint any country in the world on a map.

Primal Triggers vs. Reflection

Our brains are wired for stories and games. The thrill of the hunt, the anticipation of uncovering the treasure, the challenge of competition or making it past the next stage -- all of these desires drive the motivational synapses that make learning happen at a rate far surpassing the traditional, unmotivated, context-less content that so many students and working adults face daily.

This begs the question: if games can teach us more quickly and more deeply, why don't we just dedicate a majority of our educational resources to the development of quality games?

Of course, before we jump into this too far, we might want to look for some proof that those kids who spend hours every day in the gamers' trance are actually getting smarter and will turn out to be innovative world-changers.

Problem is . . . most of them aren't.

The reality is that, while the brain may be firing on all cylinders as part of playing a game, learning which can be used in life requires a higher process still. It requires reflection.

And herein lies the rub which prevents many video games from becoming true teaching tools. The very thing that makes them so compelling is also their Achilles heel.

Games don't just capture our imaginations; they tend to hijack them. Game worlds and stories are so immersive, so compelling that their passengers lose the desire to get off the ride. It's the same tool that authors use at the end of their chapters to leave a whole class of sixth-graders groaning and wanting to know what comes next. Except in the game world, it's amplified umpteen times.

The problem is, there is no exit ramp, no good launch point for saying, "Now that you've learned this, go apply it in a different context, go create something, go explore, go teach, go dream." So the brain just keeps cycling over the same story and over the same problems, unable to release them productively.

Like a rocket held in orbit by the planet's gravitational pull, it takes a tremendous amount of force and effort to reach escape velocity. And for the most part, gamers don't escape the pull.

Level Up

The truth is, though, it is possible to build escape routes into game mechanics. It's possible to establish breakpoints that don't just recognize achievement but propel users to outside thinking and activity.

It's possible that the incredible potential force built up by games could be unleashed to actually make a difference in the real world, but it would require designers adopting a different model which doesn't equate design success with addictiveness. With any luck though, the coming years will see a plethora of new ideas, platforms and games which are more narrative, thoughtful and encouraging of reflection than those whose primary goal is mass captivation without a break. It may be then that we finally enter the realm of the game not only as a learning tool, but as an art form.

Then perhaps there will be less of parents playing this generational tug-of-war, and kids will have something to be engaged in besides picking locks.